Quadratic Equations (Algebra)

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JEE Mains Problems

Q1: Let α, $\alpha + 2$, $\alpha \in Z$, be the roots of the quadratic equation

$x(x + 2) + (x + 1)(x + 3) + (x + 2)(x + 4) + \ldots\ldots\ldots + (x + n + 1)(x + n - 1) = 4n$ for some $n \in N$. Then $n + \alpha$ is equal to :

[JEE Mains 2nd April 2026 Shift 1]

A) 0

B) 1

C) 2

D) 3

{Solution: https://www.youtube.com/watch?v=$_{W}AVSUet9a0$}